Thermochimica Acta 551 (2013) 104–109

Contents lists available at SciVerse ScienceDirect

Thermochimica Acta

journal homepage: www.elsevier.com/locate/tca

Thermodynamic assessment of the La–Sb and the Ho–Sb systems using the associate model

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ARTICLE INFO

Article history:
Received 2 July 2012
Received in revised form 25 October 2012
Accepted 29 October 2012
Available online 15 November 2012

Keywords:
CALPHAD
Phase diagram
First-principles calculation
The associate model
The La–Sb system
The Ho–Sb system

ABSTRACT

The La–Sb and the Ho–Sb systems were thermodynamically assessed by CALPHAD (CALculation of PHAse Diagram) approach based on the available experimental data including thermodynamic properties and phase equilibria. The formation enthalpies (at 0 K) of the compounds of the La–Sb and the Ho–Sb systems were calculated by the first-principles method and used in the present thermodynamic optimization. The liquid phases of both systems were modeled as the associate model, with the constituent species La, Sb and La_3Sb_2 for the La–Sb liquid and Ho, Sb and HoSb for the Ho–Sb liquid. The excess Gibbs energy was formulated with the Redlich–Kister polynomial. The intermetallics, La_2Sb, La_3Sb_2, LaSb, $LaSb_2$, Ho_5Sb_3, α-Ho_4Sb_3, β-Ho_4Sb_3, α-HoSb, β-HoSb and $HoSb_2$, were treated as stoichiometric compounds. The calculated results showed the good agreement with the experimental data. The inconsistency of the calculated formation enthalpy of the compound $HoSb_2$ with the reported experimental value was analyzed.
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1. Introduction

The $CoSb_3$-based compounds have the typical skutterudite structure and show good thermoelectric properties such as low lattice thermal conductivities and high power factors due to the rattling of filling atoms such as rare earth elements (RE) added to the cage like voids in the crystal structure [1–5]. The RE–Sb systems deserve to be studied in order to accomplish the Co–RE–Sb thermodynamic database for developing high-performance alloys. The phase diagrams of the related systems may be straightforward and convenient to offer the guidance for the material design and the process determination by CALPHAD method. In order to better understand the interactions of La and Ho with Sb, thermodynamic properties and phase equilibria concerning the La–Sb and the Ho–Sb systems should be studied.

As part of the Co–RE–Sb thermodynamic database, this work presents the thermodynamic descriptions of the La–Sb and the Ho–Sb binary systems based on the available experimental data and the first-principles calculation.

2. Evaluation of the available information

2.1. The La–Sb system

The phase equilibria of the La–Sb binary system were previously measured by Vogel and Klose [6]. It was reported that four intermetallics, La_2Sb, La_3Sb_2, LaSb, $LaSb_2$, exist in this system. The La_2Sb, LaSb, and $LaSb_2$ phases are formed from the peritectic reactions, $L + La_3Sb_2 \leftrightarrow La_2Sb$ at 1733 K, $L + La_3Sb_2 \leftrightarrow LaSb$ at 1813 K, and $L + LaSb \leftrightarrow LaSb_2$ at 1383 K, respectively, whereas the La_3Sb_2 phase melts congruently at about 1963 K. In view of the fact that the solubility of each intermetallic compound is very limited, they were treated as stoichiometric compounds in the present work. There are two eutectic reactions, $L \leftrightarrow$ Fcc_A1 (La) + La_2Sb at 1118 K and $L \leftrightarrow$ Rhomb (Sb) + $LaSb_2$ at 903 K. Because the terminal phases in both sides have very narrow homogeneity ranges, they are treated as pure element phases in the present assessment. Furthermore, a polymorphic transformation, Fcc_A1 (La) ↔ Bcc_A2 (La) at 1138 K, was observed. The details of the invariant reactions in this system are shown in Table 1.

Colinet et al. [7] and Takeuchi and Inoue [8] calculated the value of the enthalpy of mixing of the liquid phase at 50 at.% La. The formation enthalpies of the intermetallic compounds, La_2Sb, LaSb and $LaSb_2$, were measured in Refs. [9–11] by calorimetry and EMF methods, and they were predicted in Ref. [7] by Miedema model.

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http://dx.doi.org/10.1016/j.tca.2012.10.026

Table 1
Invariant reactions of the La–Sb binary system.

Invariant reaction	Reaction type	T (K)	x^L_{Sb}	Reference
L ↔ Fcc_A1 (La) + La_2Sb	Eutectic	1118	~0.034	[6]
		1117	0.04	This work
L + La_3Sb_2 ↔ La_2Sb	Peritectic	1733	~0.324	[6]
		1733	0.31	This work
L + La_3Sb_2 ↔ LaSb	Peritectic	1813	~0.53	[6]
		1813	0.522	This work
L ↔ La_3Sb_2	Congruent	1963	0.4	[6]
		1963	0.4	This work
L + LaSb ↔ $LaSb_2$	Peritectic	1383	~0.745	[6]
		1383	0.72	This work
L ↔ Rhomb(Sb) + $LaSb_2$	Eutectic	903	~0.995	[6]
		902.5	0.996	This work

2.2. The Ho–Sb system

The phase equilibria of the Ho–Sb binary system were previously measured by Abdusalyamova et al. [12] based on differential thermal analysis (DTA), metallography and X-ray methods. It was reported that six intermetallics, Ho_5Sb_3, α-Ho_4Sb_3, β-Ho_4Sb_3, α-HoSb, β-HoSb, and $HoSb_2$, and two terminal solution phases, Hcp_A3 (Ho) and Rhomb_A7 (Sb), exist in this system. The compounds Ho_5Sb_3, β-Ho_4Sb_3, and $HoSb_2$ are formed by the peritectic reactions, L + β-Ho_4Sb_3 ↔ Ho_5Sb_3 at 1953 K, L + α-HoSb ↔ β-Ho_4Sb_3 at 2103 K, and L + α-HoSb ↔ $HoSb_2$ at 943 K, respectively. The Ho_4Sb_3 has a reversible polymorphic transition β-Ho_4Sb_3 ↔ α-Ho_4Sb_3 at 1893 K, and the latter decomposes by the eutectoid reaction, α-Ho_4Sb_3 ↔ α-HoSb + Ho_5Sb_3 at 1723 ± 100 K. The HoSb phase melts congruently at about 2433 K, and it has a polymorphic transformation β-HoSb ↔ α-HoSb at a temperature in the range of 2123–2373 K. Abdusalyamova et al. [12] and Abdusalyamova [13] also reported the more accurate data for α-Ho_4Sb_3 ↔ α-HoSb + Ho_5Sb_3 at 1723 ± 50 K and β-HoSb ↔ α-HoSb at 2153 K. Furthermore, the polymorphic transformation, Bcc_A2 (Ho) ↔ Hcp_A3 (Ho) at 1698 K, was observed. The details of the invariant reaction in this system are shown in Table 2.

In addition, the thermochemical data of the Ho–Sb binary system were reported by several groups. The enthalpy of mixing of the liquid phase at 50 at.% Ho were calculated in Refs. [7,8]. Using electromotive force (EMF) measurements, Goryacheva et al. [14] reported the enthalpies of formation of $HoSb_2$ and HoSb phases with respect to Hcp_A3 (Ho) and Rhombohedral_A7 (Sb). Pratt and Chua [15] measured the enthalpy of formation of the HoSb phase. Colinet et al. [7] predicted the values of the formation enthalpies of the intermetallic compounds Ho_5Sb_3, HoSb and $HoSb_2$ by Miedema model.

3. Calculation method

3.1. First-principle calculations

In order to examine the experiment data of thermochemical properties for intermetallics and have a more reliable description for the thermodynamic parameters, the first-principles method was used to calculate the formation enthalpies of the binary compounds in the La–Sb and the Ho–Sb systems. The scalar relativistic all-electron Blöchl's projector augmented-wave (PAW) [16,17] method within the generalized gradient approximation (GGA) was adopted, as implemented in the highly efficient Vienna ab initio simulation package (VASP) [18,19]. For the GGA exchange–correlation functional, the Perdew–Burke–Ernzerhof parameterization (PBE) [20–22] was employed, which is not beyond GGA. A plane-wave energy cutoff of 400 eV was held constant for the A_xB_y compounds (A = La or Ho, B = Sb, and x and y are the stoichiometric numbers of the elements A and B, respectively, in the related compound), and Brillouin zone integrations were performed using Monkhorst and Pack [23] and Methfessel and Paxton [24] K-point meshes, and the K-point meshes of compounds were 12 × 12 × 12. The total energy was converged numerically to less than 1×10^{-6} eV/unit with respect to electronic, ionic and unit cell degrees of freedom. All calculations were performed using the "high" setting within VASP to avoid wrap-around errors. Spin polarization was used in all calculations.

The formation enthalpies of the intermetallic compounds are calculated by the following equation:

$$\Delta H(A_xB_y) = E_{\text{total}}(A_xB_y) - xE_{\text{total}}(A) - yE_{\text{total}}(B) \quad (1)$$

where E_{total} (A_xB_y), E_{total} (A), and E_{total} (B) are the calculated total energies (per atom at T = 0 K) of the intermetallic compounds, pure A and B, respectively.

3.2. Thermodynamic modeling

The lattice stabilities of all pure elements are cited from Dinsdale [25]. Different models are employed to describe the liquid phase and intermetallic compounds in the La–Sb and the Ho–Sb binary systems.

Table 2
Invariant reactions of the Ho–Sb binary system.

Invariant reaction	Reaction type	T (K)	x^L_{Sb}	**Reference**
L ↔ Ho_5Sb_3 + Hcp_A3 (Ho)	Eutectic	1443	0.15	[12]
		1443	0.149	This work
L + β-Ho_4Sb_3 ↔ Ho_5Sb_3	Peritectic	1953	~0.317	[12]
		1953	0.31	This work
L + α-HoSb ↔ β-Ho_4Sb_3	Peritectic	2103	~0.36	[12]
		2103	0.35	This work
L ↔ β-HoSb	Congruent	2433	0.5	[12]
		2434	0.5	This work
L + α-HoSb ↔ $HoSb_2$	Peritectic	943	~0.98	[12]
		943	0.977	This work
L ↔ $HoSb_2$ + Rhomb_A7 (Sb)	Eutectic	897	~0.99	[12]
		895.6	0.984	This work
β-Ho_4Sb_3 ↔ α-Ho_4Sb_3	Polymorphic	1893	–	[12]
		1893	–	This work
α-Ho_4Sb_3 ↔ α-HoSb + Ho_5Sb_3	Eutectoid	1723 ± 100	–	[12]
		1723 ± 50	–	[13]
		1722	–	This work
β-HoSb ↔ α-HoSb	Polymorphic	2123–2373	–	[12]
		2153	–	[13]
		2153	–	This work

3.2.1. Liquid phase

The liquid phase is described with the associate model [26] to describe the asymmetry liquid phase boundary lines and deviated points of invariant reactions reasonably. The associate species La_3Sb_2 and HoSb are considered in the liquid phase of the La–Sb and the Ho–Sb systems, respectively, and each of them corresponds to the compound with the highest congruent melting point in the related system. The Gibbs energy of liquid phase is expressed with the following expression [27]:

$$\begin{aligned} G_m^{\mathrm{Liq}} &= x_A^0 G_A^{\mathrm{Liq}} + x_B^0 G_B^{\mathrm{Liq}} + x_{A_xB_y}^0 G_{A_xB_y}^{\mathrm{Liq}} \\ &\quad + RT(x_A \ln x_A + x_B \ln x_B + x_{A_xB_y} \ln x_{A_xB_y}) \\ &\quad + x_A x_B \sum_{i=0}^{n} {}^iL_{A,B}^{\mathrm{Liq}} (x_A - x_B)^i + x_A x_{A_xB_y} \sum_{i=0}^{n} {}^iL_{A,A_xB_y}^{\mathrm{Liq}} (x_A - x_{A_xB_y})^i \\ &\quad + x_{A_xB_y} x_B \sum_{i=0}^{n} {}^iL_{A_xB_y,B}^{\mathrm{Liq}} (x_{A_xB_y} - x_B)^i \end{aligned} \tag{2}$$

where R is the gas constant, x_i is the mole fraction of the component i, A_xB_y is the considerable ordering of A and B at the $x/(x+y)$ liquid composition, and ${}^iL_{m,n}^{\mathrm{Liq}}$ $(i = 0, 1, 2;\ m, n = A, B, A_xB_y;\ m \neq n)$ is the interaction parameter between species m and n, which is formulated as a linear function of temperature as ${}^iL_{m,n}^{\mathrm{Liq}} = a + bT$, and the coefficients a and b are the parameters to be optimized.

3.2.2. Intermetallic compounds

There are totally 10 intermetallic compounds in the La–Sb and the Ho–Sb binary systems. All these compounds, La_2Sb, La_3Sb_2, LaSb, $LaSb_2$, α-Ho_5Sb_3, Ho_4Sb_3, β-Ho_4Sb_3, α-HoSb, β-HoSb and $HoSb_2$, are treated as stoichiometric compounds A_mB_n because the homogeneity ranges of the compounds are very narrow. Meanwhile, due to lack of heat capacities of these phases and with the Neumann–Kopp rule, Gibbs energy per mole of atoms is expressed as follows:

$$G_m^{A_mB_n} = \frac{m}{m+n} {}^0G_A^{\mathrm{SER}} + \frac{n}{m+n} {}^0G_B^{\mathrm{SER}} + c + dT \tag{3}$$

where the parameters c and d are to be optimized.

4. Results and discussion

4.1. First-principles calculations

By the first-principles method, the formation enthalpies of the binary compounds in the La–Sb and the Ho–Sb systems were calculated. The comparison between the calculated results and the experimental data is listed in Table 3. For the La–Sb system, the results of first-principles calculations are consistent with the data reported by Borzone et al. [11], with the relative deviations within ±10%. And for the Ho–Sb system, some problems exist. The calculation result of HoSb phase is consistent with the experimental data [14], with the relative deviations about ±14%. But the formation enthalpy of $HoSb_2$ phase was calculated to be −19.679 kJ/mol, which is much different from the reported experimental data [14], −84.4 kJ/mol. And the value, −19.679 kJ/mol, is too small to the $HoSb_2$ phase existing in this system. Therefore, it is doubted that the reported crystal data is reasonable. To verify this supposition, the formation enthalpy of $HoSb_2$ was calculated by first-principles calculation using the structures of $EuSb_2$, $YbSb_2$ and $SmSb_2$, since Eu, Yb and Sm are the rare earth elements similar as Ho. By using the structures of $EuSb_2$, $YbSb_2$ and $SmSb_2$, the formation enthalpy of $HoSb_2$ were calculated to be −64.546 kJ/mol, −65.551 kJ/mol and −69.418 kJ/mol, respectively, which are more negative than the result calculated by using the reported structure of $HoSb_2$. This means that $HoSb_2$ would be more stable when taking the structures of $EuSb_2$, $YbSb_2$ and $SmSb_2$. Therefore, the structure of $HoSb_2$ needs to be further determined experimentally. The information of crystal structure and relative calculated formation enthalpies are listed in Table 3.

In sum, the formation enthalpies of La_2Sb, LaSb, Ho_5Sb_3 and HoSb phases were calculated by first-principles methods, and the calculation results of La_2Sb, LaSb and HoSb demonstrate the closer consistent with the experimental data from Refs. [11,14] than the reported data from Refs. [7,9,10,15]. Because of the absence of experiment data, the formation enthalpy of Ho_5Sb_3 is predicted by first-principle calculations to assist the thermodynamic calculation. Therefore, the formation enthalpies calculated by first-principles methods and experiment data from Refs. [11,14] are adopted in the present work during the assessments of the La–Sb and the Ho–Sb binary systems.

Table 3
The formation enthalpies at 0 K obtained by the first-principles calculations, the data reported by literature and the value calculated by Miedema model.

Phase	Pearson symbol	Space group	Prototype	$\Delta H_{\mathrm{formation}}$ (J/mol)	Reference
La_2Sb	tI12	I4/mmm	La_2Sb	−88,090	First-principle
	–	–	–	−97,900	Experiment [11]
	–	–	–	−142,089	Experiment [9]
	–	–	–	−94,000	Miedema [7]
LaSb	cF8	Fm-3m	NaCl	−123,511	First-principle
	–	–	–	−130,500	Experiment [11]
	–	–	–	−148,197	Experiment [10]
	–	–	–	−118,000	Miedema [7]
Ho_5Sb_3	hP16	P63/mcm	Mn_5Si_3	−88,488	First-principle
	–	–	–	−82,000	Miedema [7]
HoSb	cF8	Fm-3m	NaCl	−108,284	First-principle
	–	–	–	−126,500	Experiment [14]
	–	–	–	−64,000	Experiment [15]
	–	–	–	−91,000	Miedema [7]
$HoSb_2$	oS6	C222	$HoSb_2$	−19,679	First-principle
	mP6	P21/m	$CaSb_2$ (from $EuSb_2$)	−64,546	First-principle
	oS12	Cmcm	$ZrSi_2$ (from $YbSb_2$)	−64,551	First-principle
	oS24	Cmca	$SmSb_2$ (from $SmSb_2$)	−69,418	First-principle
	–	–	–	−84,400	Experiment [14]
	–	–	–	−74,000	Miedema [7]

Table 4
Thermodynamic parameters of the La–Sb ternary system obtained in this work.

Phase	Thermodynamic parameters
Liquid (La, La_3Sb_2, Sb)	$G^{Liq}_{La_3Sb_2} = -399,723.017 + GHSERLA + GHSERSB$
	${}^0L^{Liq}_{La,Sb} = -292,000$
	${}^1L^{Liq}_{La,Sb} = -16,000$
	${}^2L^{Liq}_{La,Sb} = 66,000$
	${}^0L^{Liq}_{La,La_3Sb_2} = -94,973.4386$
La_2Sb	$G^{La_2Sb}_{La:Sb} = -97,000 + 4.813T + 0.667GHSERLA + 0.333GHSERSB$
La_3Sb_2	$G^{La_3Sb_2}_{La:Sb} = -114,000 + 7.932T + 0.6GHSERLA + 0.4GHSERSB$
LaSb	$G^{LaSb}_{La:Sb} = -121,800 + 13.317T + 0.5GHSERLA + 0.5GHSERSB$
$LaSb_2$	$G^{LaSb_2}_{La:Sb} = -93,250 + 10.977T + 0.333GHSERLA + 0.667GHSERSB$

4.2. Thermodynamic assessment

With the lattice stabilities of all pure elements cited from Dinsdale [25] and the PARROT module of the Thermo-Calc software [28], the phase relations in the La–Sb and the Ho–Sb binary systems were carefully assessed based on the experimental and theoretical data of the phase diagram and thermodynamic properties. Tables 4 and 5 list the optimized thermodynamic parameters for the La–Sb and the Ho–Sb systems, respectively. As shown in the next sections, most of the thermodynamic and phase boundary data are well reproduced.

4.2.1. The La–Sb binary system

Fig. 1 shows the calculated phase diagram of the La–Sb system together with the experimental data taken from Ref. [6]. It can be seen that the calculated phase diagram is in reasonable agreement with the experimental data. The calculated peritectic temperatures of the intermetallic compounds, La_2Sb, LaSb, and $LaSb_2$, are 1733, 1813 and 1383 K, respectively, the congruent melting temperature of La_3Sb_2 is 1963 K, and the eutectic reactions, L ↔ Fcc_A1 (La) + La_2Sb and L ↔ Rhomb (Sb) + $LaSb_2$, are 1117 and 902.5 K. The details of the invariant reactions in this binary system together with the data from the literatures are listed in Table 1 for comparison.

Fig. 2 presents the calculated mixing enthalpies of the liquid phase at 2500 K, together with the results reported by Refs. [7,8], from which it can be seen that a good agreement is obtained. Fig. 3 represents the calculated formation enthalpies of intermetallic compounds at 300 K together with the experiment data [9–11] and the first-principles calculations. The calculated values agree very well with the results by the present first-principles calculations and the data reported by Borzone et al. [11].

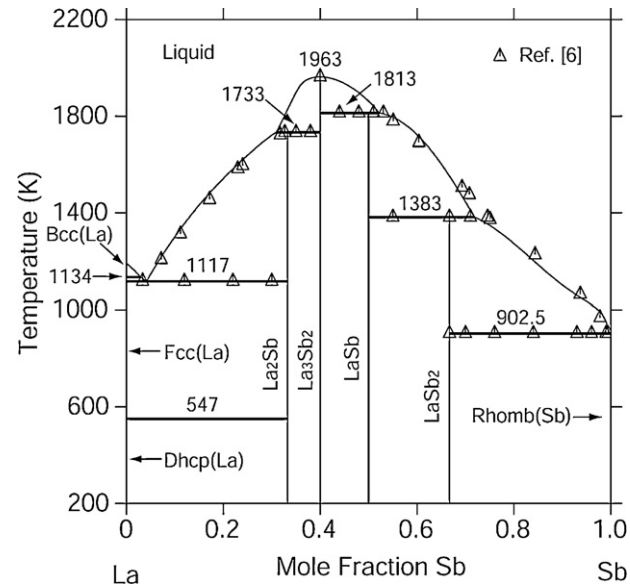

Fig. 1. The assessed La–Sb phase diagram in comparison with the experimental data [6].

4.2.2. The Ho–Sb binary system

The calculated phase diagram is shown in Fig. 4 together with the experimental data from Abdusalyamova et al. [12] and Abdusalyamova [13]. Two reversible polymorphic transitions β-HoSb ↔ α-HoSb and β-Ho_4Sb_3 ↔ α-Ho_4Sb_3 were, respectively, obtained at 2153 and 1893 K in the calculated results, the peritectic temperatures of the intermetallic compounds, Ho_5Sb_3, β-Ho_4Sb_3, and $HoSb_2$, are 1953, 2103, and 943 K, respectively, and the congruent melting temperature of β-HoSb is 2434 K, the eutectic reactions, L ↔ Hcp_A3 (Ho) + Ho_5Sb_3 and L ↔ Rhomb (Sb) + $HoSb_2$, are 1443 and 895.6 K, respectively, and the eutectoid reaction,

Table 5
Thermodynamic parameters of the Ho–Sb ternary system obtained in this work.

Phase	Thermodynamic parameters
Liquid (Ho, HoSb, Sb)	$G^{Liq}_{HoSb} = -107,771.083 + 3.896T + GHSERHO + GHSERSB$
	${}^0L^{Liq}_{Ho,Sb} = -234,000 - 13.5T$
	${}^1L^{Liq}_{Ho,Sb} = -10,000 - 14T$
	${}^2L^{Liq}_{Ho,Sb} = 64,637.5624 + 2.361T$
	${}^3L^{Liq}_{Ho,Sb} = 53,407.5731$
	${}^0L^{Liq}_{Ho,HoSb} = -181,998.143 + 5T$
	${}^1L^{Liq}_{Ho,Sb} = 58,889.3986 + 6.963T$
Ho_5Sb_3	$G^{Ho_5Sb_3}_{Ho,Sb} = -97,500 + 5.942T + 0.625GHSERHO + 0.375GHSERSB$
β-Ho_4Sb_3	$G^{\beta\text{-}Ho_4Sb_3}_{Ho,Sb} = -100,950 + 3.4446T + 0.5714GHSERHO + 0.4286GHSERSB$
α-Ho_4Sb_3	$G^{\alpha\text{-}Ho_4Sb_3}_{Ho,Sb} = -101,650 + 3.8145T + 0.5714GHSERHO + 0.4286GHSERSB$
β-HoSb	$G^{\beta\text{-}HoSb}_{Ho:Sb} = -109,037.263 + 3.2935T + 0.5GHSERHO + 0.5GHSERSB$
α-HoSb	$G^{\alpha\text{-}HoSb}_{Ho:Sb} = -119,662.836 + 8.2287T + 0.5GHSERHO + 0.5GHSERSB$
$HoSb_2$	$G^{HoSb_2}_{Ho:Sb} = -79,650 + 5.0255T + 0.333GHSERHO + 0.667GHSERSB$

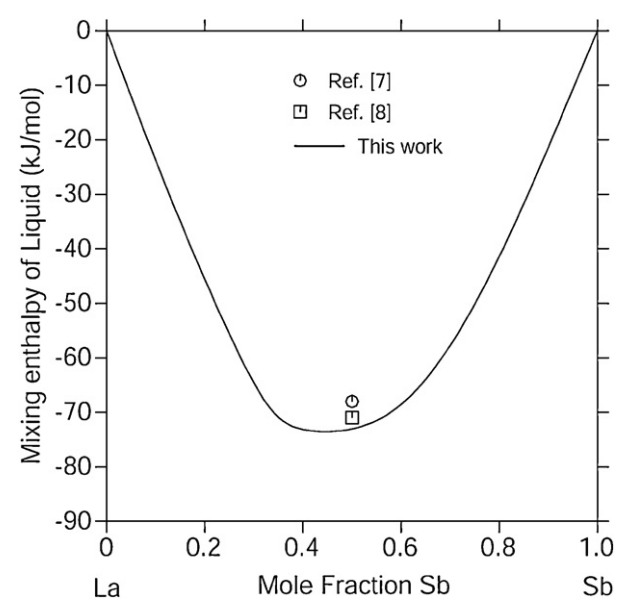

Fig. 2. The mixing enthalpy of liquid La–Sb alloys at 2500 K: comparison between calculated values and reported data [7,8]. Reference states are liquid La and liquid Sb.

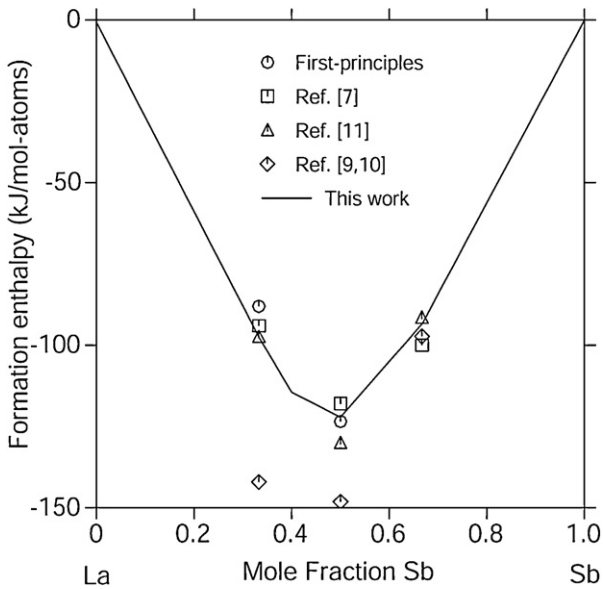

Fig. 3. The formation enthalpy of La–Sb intermetallic compounds at 300 K: comparison between calculated values with reported data [7,9–11]. Reference states are Dhcp (La) and Rhomb (Sb).

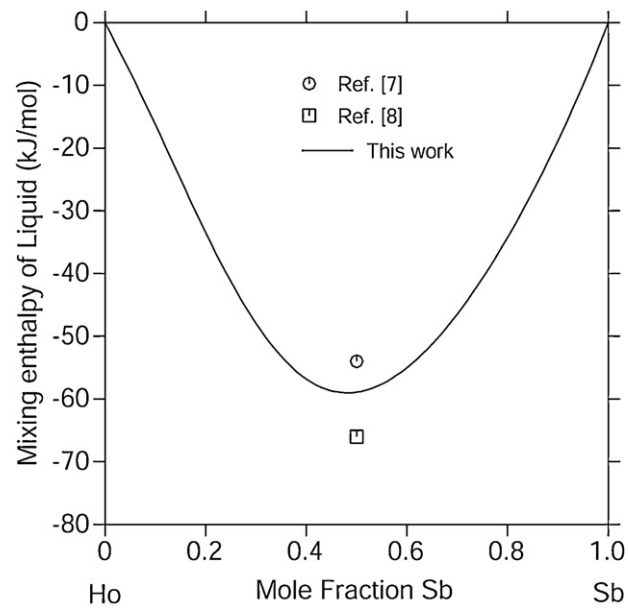

Fig. 5. The mixing enthalpy of liquid Ho–Sb alloys at 3000 K: comparison between calculated values and reported data [7,8]. Reference states are liquid Ho and liquid Sb.

α-$Ho_4Sb_3 \leftrightarrow \alpha$-HoSb + Ho_5Sb_3, is 1723 K. All the calculated invariant reactions in the Ho–Sb system are listed in Table 2. The calculated results are in good agreement with most of the experimental data [12,13].

Fig. 5 presents the calculated mixing enthalpies of the liquid phase at 3000 K, together with the results reported by Refs. [7,8], from which it can be seen that a good agreement is obtained. The calculated formation enthalpies of the intermetallic compounds at 773 K are compared with the experimental data [14,15] and the first-principles calculations, as shown in Fig. 6. The calculated values agree very well with the results by the present first-principles calculations and the data reported by Goryacheva et al. [14].

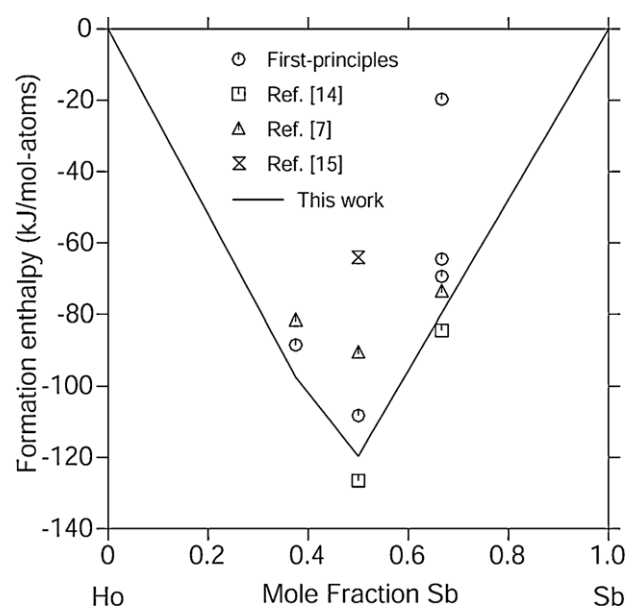

Fig. 6. The formation enthalpy of Ho–Sb intermetallic compounds at 733 K: comparison between calculated values with reported data [7,14,15]. Reference states are Hcp (Ho) and Rhomb (Sb).

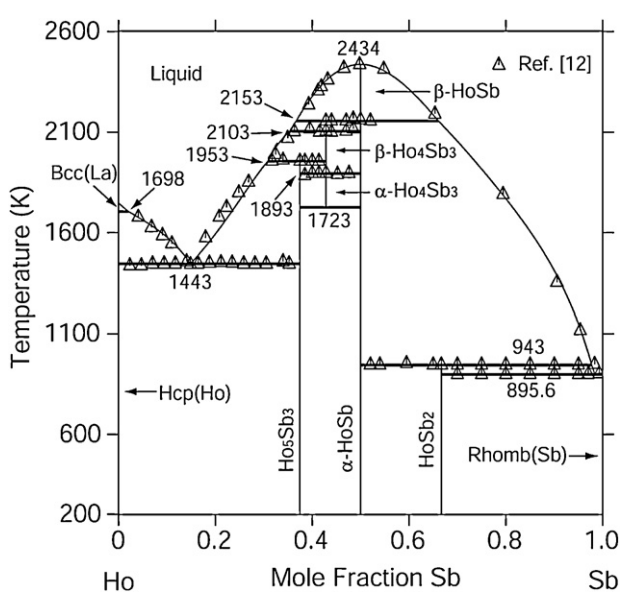

Fig. 4. The calculated Ho–Sb phase diagram in comparison with the experimental data [12].

5. Conclusions

With the assistance of the present first-principles calculations, and the experimental data of the phase diagrams and the thermodynamic properties available in literatures, the La–Sb and the Ho–Sb systems were thermodynamically assessed critically. For each of the systems, a consistent set of thermodynamic parameters for various thermodynamic models has been obtained, and all of the calculated temperatures and compositions of the invariant reactions as well as the thermodynamic properties are in good agreement with the experimental data. Especially, by using the associate model for liquid, the calculated liquidus lines in both of the La–Sb and the Ho–Sb binary systems are reasonably

reproduced. In addition, according to the inconsistency of the first-principles calculation with the reported experimental value for the formation enthalpy of the compound $HoSb_2$, it is proposed that the structure of $HoSb_2$ needs to be further determined experimentally.

Acknowledgements

This work was financially supported by National Science Foundation of China (Grant Nos. 50731002 and 50671009). Special acknowledgements were given to the Thermo-Calc software from Thermo-Calc AB, Stockholm Technology Park, Sweden, and Pandat program licensed from The CompuThermo, LLC, Madison, WI, USA.

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